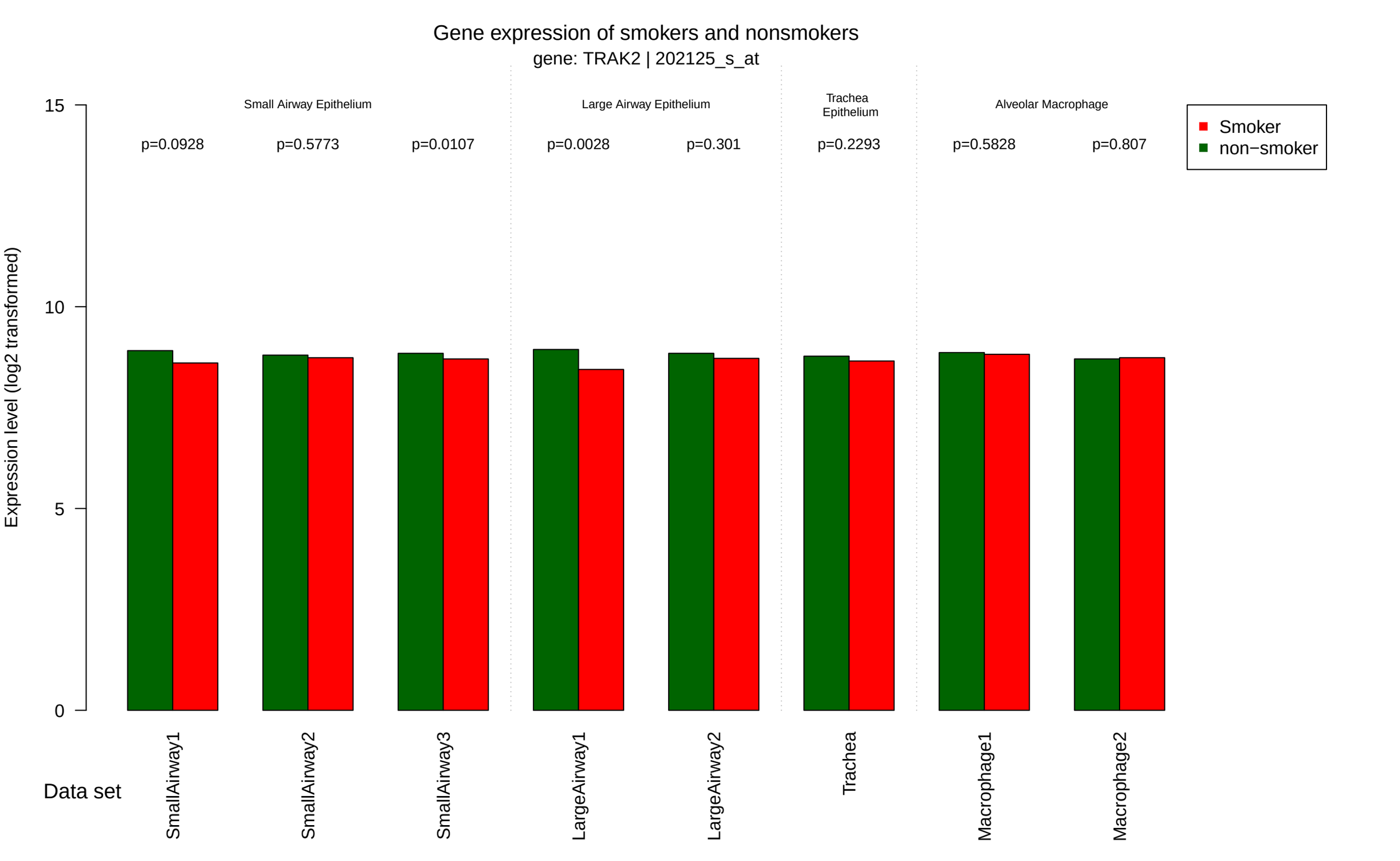
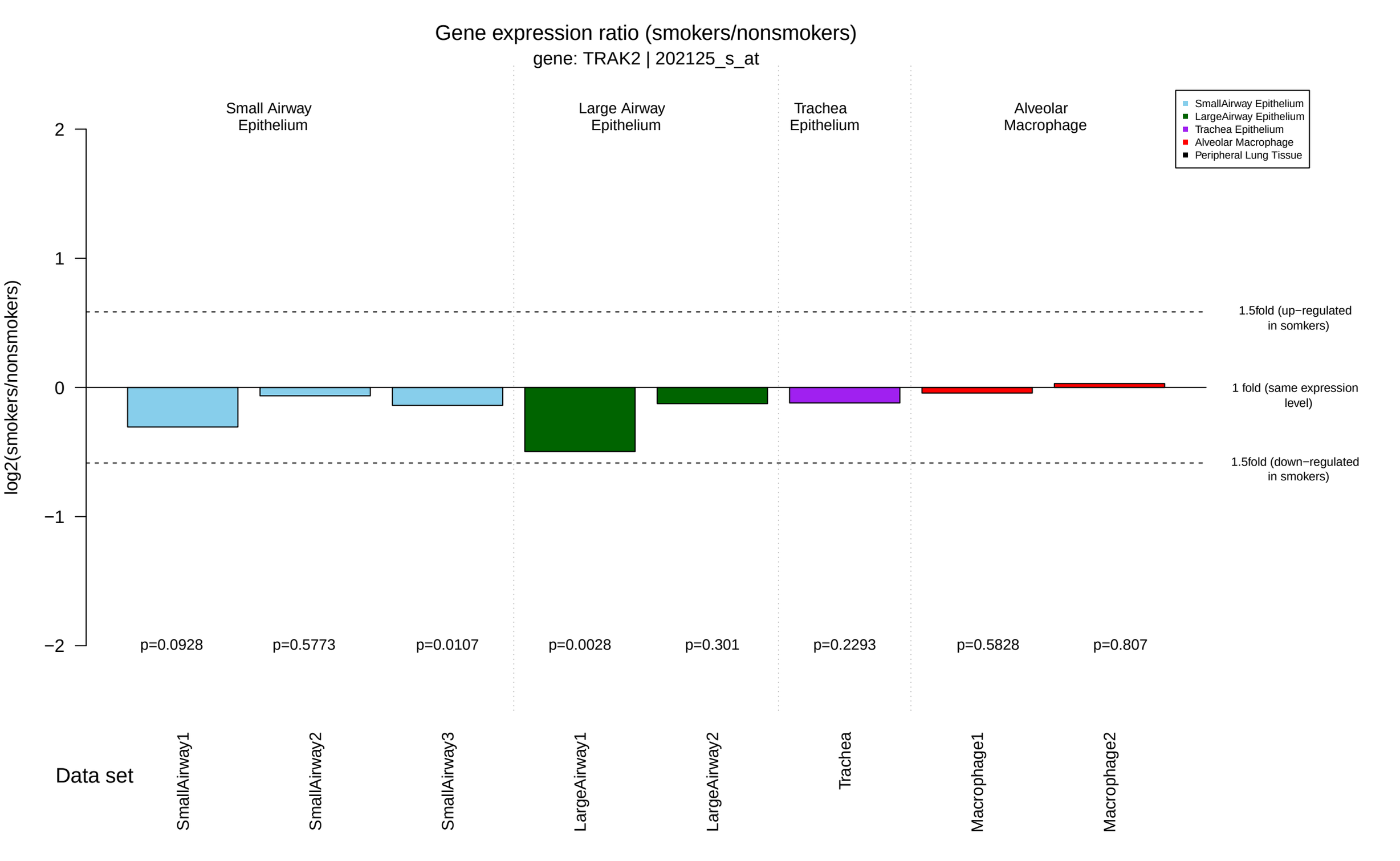
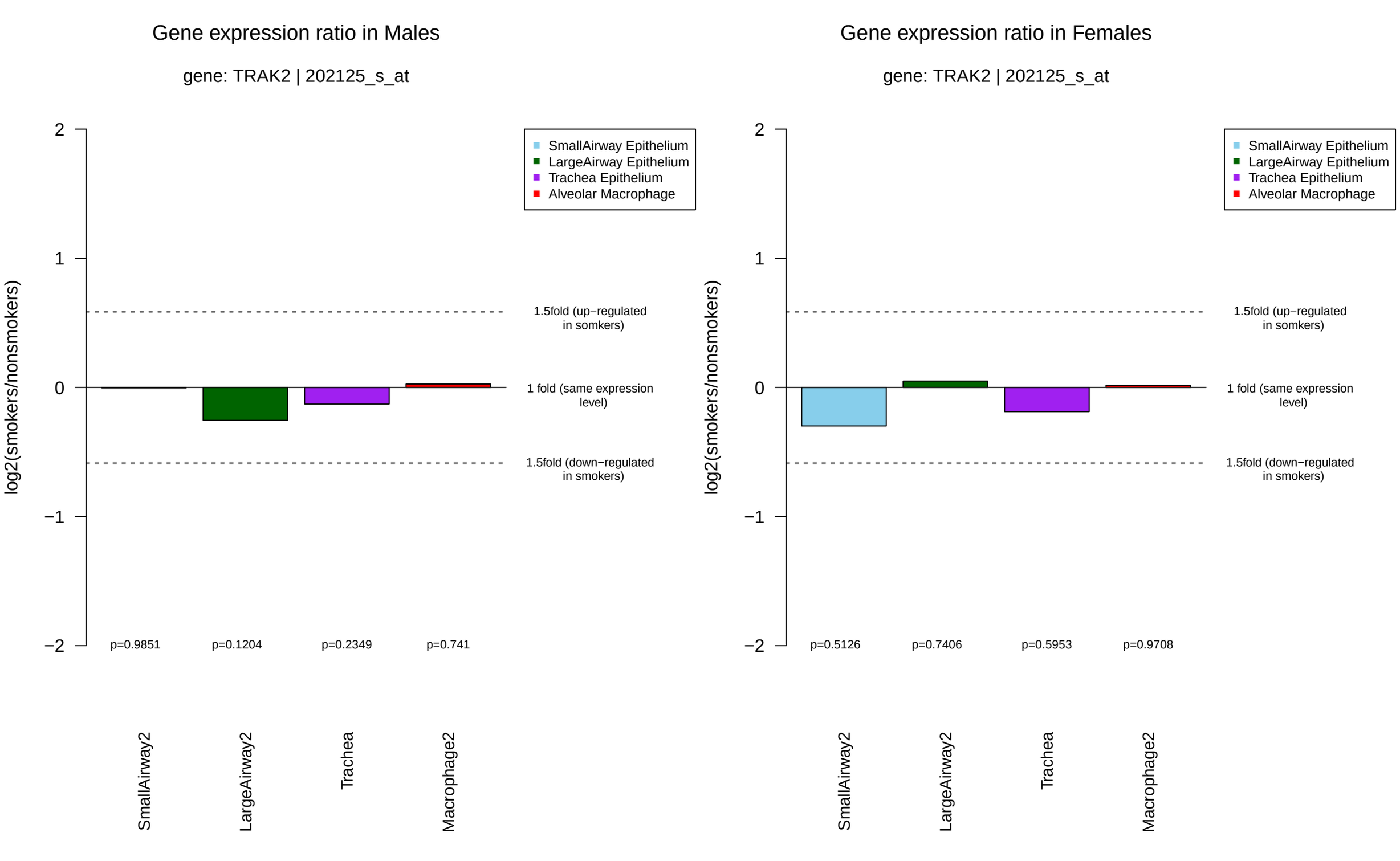
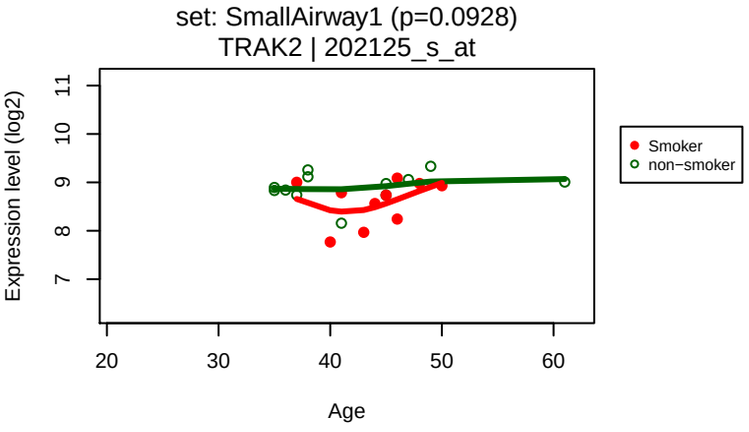
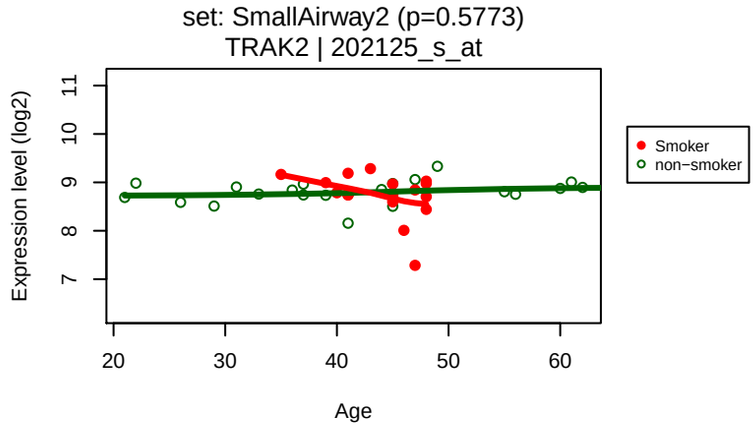
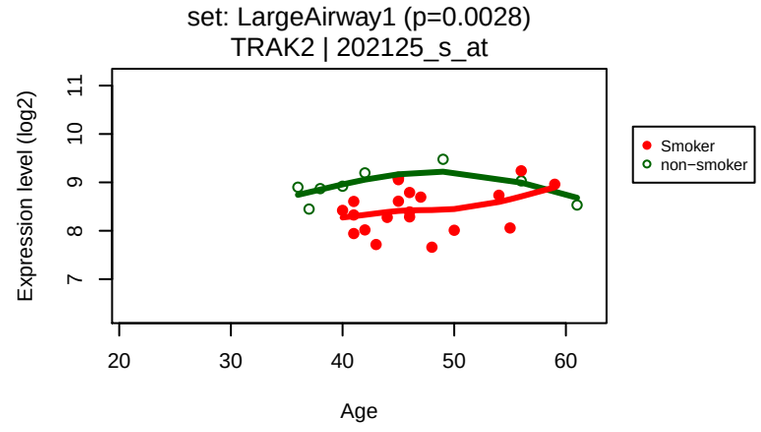

Gene expression of smokers and nonsmokers
gene: TRAK2 | 202125_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0928
p=0.5773
p=0.0107
p=0.0028
p=0.301
p=0.2293
p=0.5828
p=0.807
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

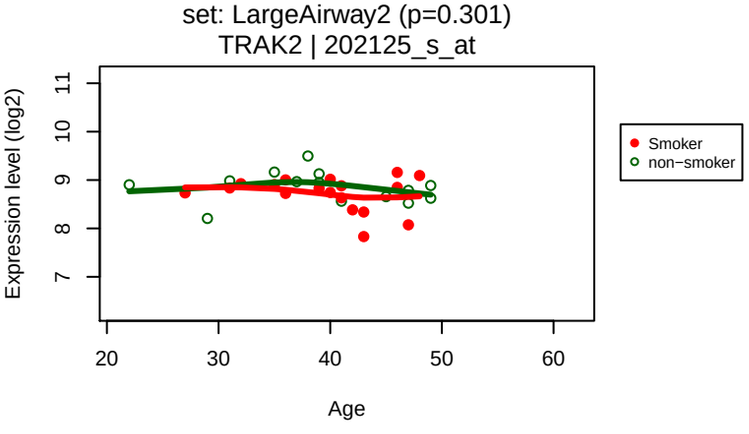

Gene expression ratio (smokers/nonsmokers)
gene: TRAK2 | 202125_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0928
p=0.5773
p=0.0107
p=0.0028
p=0.301
p=0.2293
p=0.5828
p=0.807
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

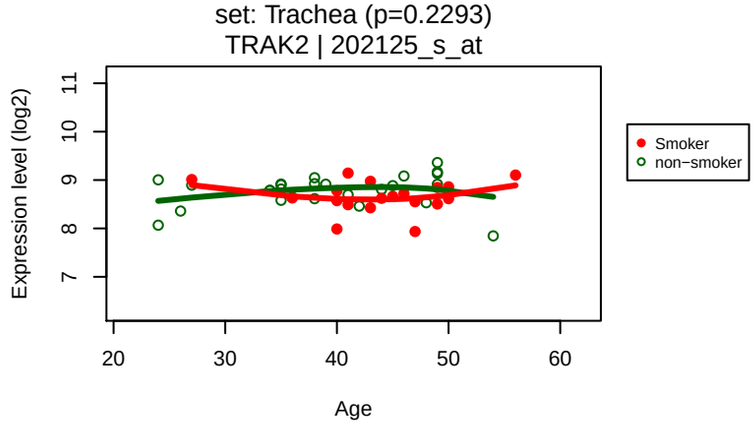

Gene expression ratio in Males
gene: TRAK2 | 202125_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9851
p=0.1204
p=0.2349
p=0.741
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK2 | 202125_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5126
p=0.7406
p=0.5953
p=0.9708
SmallAirway2
LargeAirway2
Trachea
Macrophage2

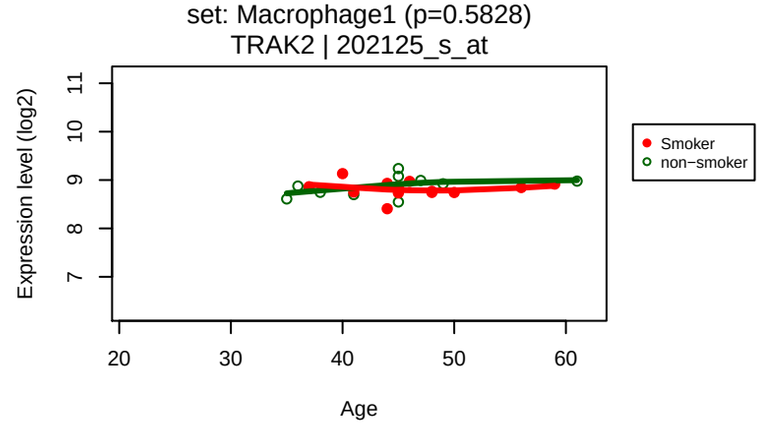

set: SmallAirway1 (p=0.0928)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker

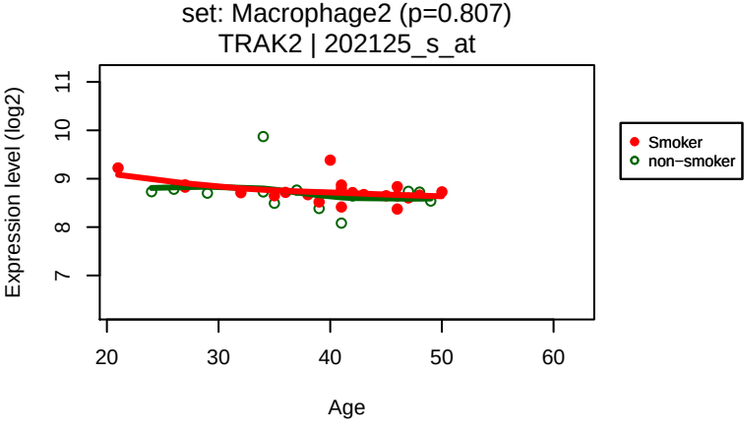

set: SmallAirway2 (p=0.5773)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker

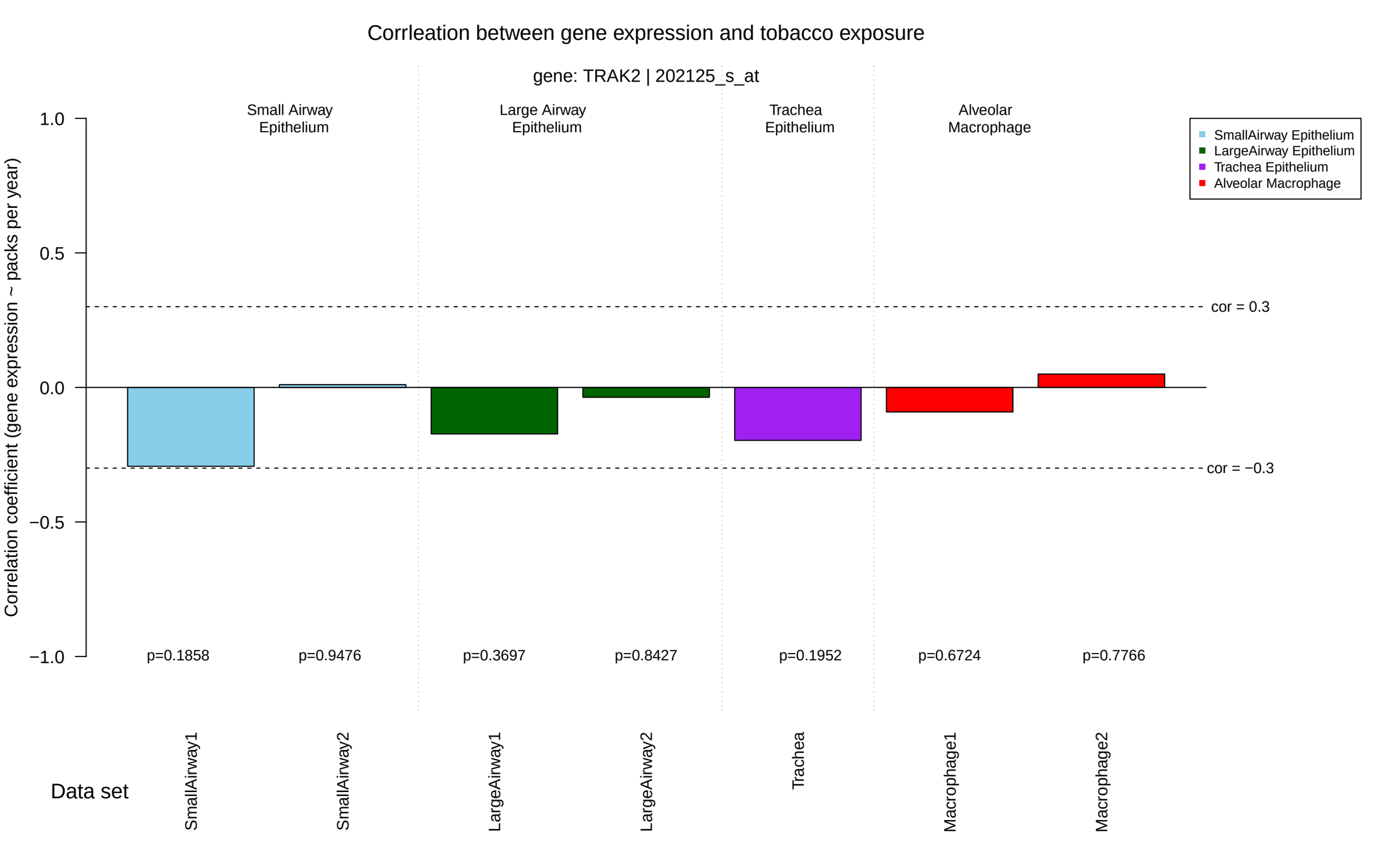

set: LargeAirway1 (p=0.0028)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.301)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2293)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5828)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.807)
TRAK2 | 202125_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK2 | 202125_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1858
p=0.9476
p=0.3697
p=0.8427
p=0.1952
p=0.6724
p=0.7766
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2